

CURRICULUM VITAE

PART 1: PERSONAL DETAILS

Name	P.M Ts Dr Hairi Zamzuri Ph.D (Control and AI Engineering)	
Current Post	CEO eMooVit Technology Sdn Bhd Futurise Center, Cyberjaya, Selangor, MALAYSIA H/P: +6012-4701449 hairi@emoovit.ai	

SUMMARY

- CEO of eMooVit Technologys Sdn Bhd for 6 years
- Experience in business development and technology development for autonomous vehicle and robotics
- 20 years experiences in as associate professor in Universiti Technoogy Malaysia
- Consultation and industries advisor panel for autonomous driving for UTEM, UIA, Politeknik
- Vast experience in vehicle advanced active safety and automated driving technology.
- 5 years experiences in consultations work for Proton Sdn Bhd on advanced driving assistance system (ADAS).
- Manage a research project with total grant of RM3,000,000.00 for 5 years between 2010-2015 from various sources of fund (FRGS, ERGS, UTM GUP, Contract research) with 12 projects as a co researcher.
- Supervise 18 students (13 Phd, 5 Master) with one PhD graduated within 3 years, graduated two Master by researchers.
- Publish a cumulative impact factor journal of 15.37 from 15 ISI journals, 6 Scopus journal and more than 30 index conferences papers with 6 H-Index from ISI and Scopus.
- Patent failing of 5 Intellectual Properties (IP).

PART 2 : QUALIFICATION AND PREVIOUS EMPLOYMENT

(Academic/ Professional/ Other Qualifications obtained)

No.	Academic/Professional Qualifications	University / Institution
5	2010 Member	Board of Engineer, Malaysia
4	2008 Ph.D (Control Engineering)	Loughborough University, UK
3	2002 M. Eng Electrical (Control)	Universiti Teknologi Malaysia, Malaysia
2	1999 B.Eng Electrical (Control)	Universiti Teknologi Malaysia, Malaysia
1	1997 Dip in Electronic Engineering	Universiti Teknologi Malaysia, Malaysia

2.2 Previous Employment

2018- current	CEO, eMooVit Technology Sdn Bhd
2017-2022	Associate Professor , Universiti Teknologi Malaysia
Nov 2014-	Senior Lecturer
2017	Malaysia-Japan International Institute of Technology (MJIIT), Universiti Teknologi Malaysia

May 2011 - March 2013	Head of UTM-PROTON Satellite Research Lab, UTM-Proton Active Safety Research Lab, Proton Satellite Research Lab
June 2010 - May 2011	IT Manager, UTM Razak School of Engineering and Advanced Technology, Universiti Teknologi Malaysia, Jalan Semarak, 54100 Kuala Lumpur, Malaysia
May 2009 - present	Senior Lecturer. UTM Razak School of Engineering and Advanced Technology, Universiti Teknologi Malaysia, Jalan Semarak, 54100 Kuala Lumpur, Malaysia.
May 2001 - April 200	Lecturer. College of Science and Technology, Universiti Teknologi Malaysia, Jalan Semarak, 54100 Kuala Lumpur, Malaysia
July 2000 - May 2001	Tutor. A College of Science and Technology, Universiti Teknologi Malaysia, Jalan Semarak, 54100 Kuala Lumpur, Malaysia
Jun 99' - July 2000	Lecturer, L&G Twintech Institute of Technology, Bandar Sri Damansara, Kuala Lumpur
Jan 1997- July 1997	RnD Technician. Sapura Research, Kuala Lumpur: Developed prototype mini PABX system 210 and 206

PART 3 : TEACHING AND SUPERVISION

3.1 Courses / Subjects (from 2008 – present)

		Credit hours			Credit hours
Undergraduate	<u>Degree</u>		<u>Diploma</u>		
	Laboratories	2(2)=8	Industrial automation	2(3)=6	
	Microprocessor	3	Digital System	3	
	Embedded System Design	3	Electronic System	3	
			Microprocessor	4(3)=12	
			Control System	3(2)=6	
			Analogue Electronic	3	
			Software Engineering	3(3)=9	
	Total	14		42	
Postgraduate	Microprocessor and Peripherals	3(3)=9			
	Embedded System Design	3			
	Total	12			

3.2 Thesis / Project Level Supervision

(Doctoral Degree/ Bachelor's Degree/ Postgraduate Diploma (M.A, M.Sc and their equivalent)

Programme	Status	Main Supervisor	Co-Supervisor
Ph.D.	Graduated/viva voce	1/1	1
	On-going	11	2
Masters (by Research)	Graduated	0	2
	On-going	6	1

Ph.D (Graduate)

- 2014 Muhammad Aizzat Zakaria (881219265373 / PD113014)
The development of robust and smooth trajectory tracking for autonomous ground vehicle in outdoor environment
- 2014 (Thesis correction) Syekh Muhammad Hafiz Fahami Zainal (821224105169 / PD 113006)
Control strategies for vehicle steer-by-wire system
- 2014 Fitriani Imaduddin (co-supervisor)
Nonlinear control design magnetorheological valve

Master (Graduated)

- 2012 Muhamad Rahairi Rani (co-supervisor)
Multiobjective Optimization of Automatic Controller Design Coefficient
- 2013 (co-supervisor) Izyan Iryani binti Mohd Yazid. (850908016848 / MJ113012)
Variable semi active studies using magnetorheological fluids based damper

Ph.D. (main supervisor)

- 11 2014 (in progress) : Nurbaiti Wahid (830402105714 / PMJ143033)
Longitudinal control for vehicle platooning
- 10 2014 (in progress) : Umar Zakir Abdul Hamid (901003065335 / PMJ143004)
Longitudinal control for vehicle platooning
- 9 2014 (in progress) : Mohammad Hafez Mohd Ariffin (780421145073 / PJ133036)
Longitudinal control for vehicle platooning
- 8 2013 (in progress) : Nurul Dayana Salim (890112145624 / PJ123034)
Robust altitude control for nonlinear Hexarotor with influence of wind disturbance
- 7 2013 (in progress) : Dafizal Derawi (880829045023 / PJ 123037)
Visual Servoing control for hexarotor in outdoor environments
- 6 2013 (in progress) : Mohd Ashadi Bin Mohd Yusop (790806085349 / PJ133013)
Back Stepping Control strategy for magnetorheological Anti-lock brake system
- 5 2013 (in progress) : Mohd Shuhaibul Fadly Mansor (810530105083 / PJ23028)
A Novel electromagnetic suspension design and control
- 4 2012 (in progress) : Zulkifli Abd Kadir
Yaw disturbance rejection control for armoured vehicle using active front steering.
- 3 2012 (in progress) : Abdul Hadi Abdul Rahman
MutliDynamic Object Detection and Tracking for obstacle avoidance
- 2 2012 (in progress) : Mohd Hatta Mohammed Ariff
Hybrid braking for electric vehicle
- 1 2011 (in-progress) Noraishikin Zulkarnain (880216145638 / PD113003)
Reducing vehicle roll using active electric anti-roll bar for ground vehicle

Ph.D. (Co-supervisor)

- 2 2013 (in-progress) Izyan Iryani binti Mohd Yazid.
Semi active suspension design using magnetorheological fluids based damper
- 1 2011 (in-progress) Shahrul Hamzah Abdul Razak (850806146203 / PE103125)
Control for CNC machine

Masters (Main supervisor)

- 6 2012-current : Sarah Atifah Saruchi (890822125580 / MJ133011)
- 5 2012-current : Mohd Hazeli Rasul (881012305059 / MJ133035)
- 4 2012 -current : Ahmad Mudzakkir Ahmad Mustafa (871224055291 / MJ33004)
Wheel slip control based on CNF control algorithm
- 3 2012 - current : Saiful Bahri Samsuri (88011956243 / PJ133013)
An improve performance of Simultaneous localisation and mapping based on statistical algorithm
- 2 2012 - current : Nor Aziah Mohd Azubir
Control Strategy for Electric Vehicle Energy Efficiency
- 1 2012 - current : Mohd Afandi Mohmad Nordin
Handling performance characteristic for all steer with independent wheel drive electric vehicle

Masters (co-supervisor)

- Master (by Course)
 - 2 2011 :Project for Msc in EBM : Maswan Yusof (740331015527)
 - 1 2011 :Project for Msc in EBM : Fadzli Solehhodin (810613086447)
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3.3 Thesis/ Project Level Evaluation

(Doctoral Degree/ Bachelor's Degree/ Postgraduate Diploma (M.A, M.Sc and their equivalent)

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| 2015 | PhD Proposal Defend, Intisar Ibrahim Radwan Saeed, An Instruement to measure student readiness for Embedded System design |
| 2015 | Series of 1 st Proposal defend for Master and PhD MJIT students |

PART 4 : RESEARCH AND CONSULTATION

4.1 Research (Own/ Group)

	Total Amount	Active Grant
Principal Researcher	RM 2,441 000	RM 172 00

Co-Researcher	RM 719 900	RM 553 000
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Principle Researcher

- 8 **UTM RUG Tier 1** (vote: 05H570 RM 90,000), 2014 (24 months), *A new torque distribution control strategy for four independent in-wheel electric vehicle*
- 7 **UTM Flagship Grant** (vote: 00G64 RM 2,000,000) 2012 (36 months), *Development of MAIORA through the road hybrid vehicle*
- 6 **FRGS grant** (vote: 4F370: RM 82,000), 2013(24 months), *Investigation of robust and smooth trajectory tracking algorithm for autonomous vehicle in sharp cornering and speed varying problem,*
- 5 **ERGS grant** (vote no: 4L033 RM 50,000), 2012(24 months), completed, *Virtual steering feedback force for vehicle steer-by-wire system using artificial intelligent approach*
- 4 **Contract Research** (vote no: 7640.4C017, RM 100,000) ,2011(24 months), completed. *UTM-Proton Active Safety research for Proton Sdh. Bhd, PROTON Berhad ,*
- 3 **UTM RUG Tier 1** (vote no: 2540.02H62, RM 107,000), 2011(24 months), completed *Reducing Vehicle roll using active electric actuator anti-roll bar system for ground vehicle,*
- 2 **FRGS** (vote no : 4F053, RM 72,000), 2010(24 months), completed. *A new robust lane detection and tracking algorithm in challenging scenario,*
- 1 **UTM Short term grant** (vote no : 3P014, RM10,000), 2009(12 months), completed *Model Based anti-roll control strategies for railway vehicle,*

Co-researcher

- 12 **PRGS** (vote: 4L648, RM 152,000), 2015(24 months), in-progress, *Real Time Water Emulsion Fuel System for Industrial Burner Application*
- 11 **UTM RUG Tier 1**(vote: 10H12 RM 50,000), 2014(24 months), in-progress, *Intelligent stereo vision-Lidar System for autonomous vehicle navigation*
- 10 **UTM RUG Tier 1**(vote: 09H90 RM 50,000), 2014(24 months), in-progress, *Pedestrian Detection and tracking using fusion of multiple laser range finder*
- 9 **UTM PAS** (vote: 01K16 RM 20,000), 2015(12 months), in-progress, *Multiple simulation and modelling tools integration for mechatronic system design.*
- 8 **UTM RUG Tier 1**(vote: 06H06 RM 90,000), 2013(24 months),in-progress, *Development of a novel multistage modular magnetorheological (MR) valve.*
- 7 **FRGS** (vote: 4F388 RM 132,500), 2013 (24 months), in-progress, *Fundamental study on cuber-physical system approach for construction industry.*
- 6 **FRGS** (Vote No. 4F332, RM 116 000). 2013(24 months), in-progress, *Determination of apparent yield stress of magnetorheological fluid in annular, radial and orifice flow channel under magnetic field excitation.*
ERGS (Vote No. 4L094, RM 111 000), 2013 (24 months), *A Novel hybrid magnetorheological damper for suspension systems.*
MJIIT Grant (Vote No. 4J046, RM 190 000.), 2012 (36 months), *Design and feasibility study of an automotive anti roll bar using field-responsive fluids.*
- 5 **UTM RUG Tier 1** (Vote No. 01H20, RM 130,000). 2011 (24 months),completed. *Design of a novel magnetorheological (MR) damper using combination of squeeze and shear modes.*
- 4 **UTM Flagship** (Vote No. 01G56, RM 50,000).2014(12 months), completed, *Development and testing of magneto-rheological damper for passively controlled turbocharger (PCT) to demonstrate capability of achieving variable stiffness in enhancing PCT operation in an engine.*
- 3 **UTM Razak School Research Fund** (Vote No. 4B027, RM 50,000).2011(18 months), completed, *Design a magnetorheological (MR) fluid testing rig with two working modes.*
- 2 **UTM Flagship** (vote no: 00G20, RM 200,000),2013(24 months), completed, *Preventive maintenance and fault diagnosis on intelligent building for energy saving.*
- 1 **UTM RUG Tier 2** (vote no: 03J74, RM 40,000), 2012(24 months), *Advanced vehicle suspension system.*

4.2 Consultation work (Own/ Group- registered with UTM only)

Academic Monitoring Team member for Kolej Kemahiran Tinggi Beranang academic curriculum, 2012-2013

4.3 Intellectual Properties

PI 2014700882	A magneto rheological isolator
PI 2014700632	A method for controlling a vehicle steer by wire
PI 2014700304	A magneto rheological device
PI 2014702681	A fuel emulsification system

PART 5 : WRITING / PUBLICATION, EVALUATION WORK AND PAPER PRESENTATION

	ISI	Scopus	Scholar
H Index	6	6	7
CIF	15.37		

5.1 Publication of Academic Books

1. S.A. Mazlan and **H. Zamzuri**, Magnetorheological Fluids: Behaviour of MR fluids, ISBN 978-3-8473-4551-0, Lambert Academic Publishing, Saarbrücken: Germany, 2012.
2. **Hairi Zamzuri**, Argyrios Zolotas, Roger Goodall and Hazlina Selamat, 'Development of Linear Nine-Degree-Of-Freedom tilting Railway Vehicle Model' book chapter on 'Studies on automatic control of ground vehicles, UTM Press ISBN 978-983-52-0666-5, 2007

5.4 Article publication in Journal Impact Factor

- 14 Imaduddin. F, Mazlan, S.A, Zamzuri, H,' Design and performance analysis of a compact magnetorheological valve with multiple annular and radial gaps,' Jurnal of Intelligent Material Systems and structures, Vol.26(9), pg 1038-1049
- 13 AH Abd Rahman, **H Zamzuri**, S.A. Mazlan, M.A.A.Rahman, Y.Yamamoto, S.B.Samsuri, Dynamic Track Management in MHT for Pedestrian Tracking using laser range finder. Mathematical Problem in Engineering, Vol 2015 (IF=1.03),
- 12 A.Y.A Fatah, S.A. Mazlan, T. Koga, **H. Zamzuri**, M. Zeinali and F. Imaduddin, A Review and Modeling of Magnetorheological Valve, International Journal of Modern Physics B, Vol. 29, 2015, 1530004. (IF: 0.455)
- 11 Z.A. Kadir, S.A. Mazlan, **H. Zamzuri**, K. Hudha and N.H. Amer, Adaptive Fuzzy-PI Control for Active Front Steering System of Armoured Vehicles: Outer Loop Control Design for Firing On The Move System, Strojnicki Vestnik-Journal of Mechanical Engineering, Vol. 61, 2015, 187-195. (IF: 0.776)
- 10 IIM Yazid, SA Mazlan, T Kikuchi, **H Zamzuri**, F Imaduddin, 'Design of magnetorheological damper with a combination of shear and squeeze modes', Materials and Design, 54, 87-95, 2014, IF 1.8
- 9 M Zeineli, S.A. Mazlan, AYA Fatah, **H Zamzuri**, 'A phenomenological dynamic model of a magnetorheological damper using a neuro fuzzy system, Smart Materials and Structures, Vol 22(12), 125013, 2013. IF 1.2
- 8 F Imaduddin, S.A.Mazlan, **H. Zamzuri**, 'A design and modelling review of rotary magnetorheological damper', Materials and Design Vol 51, 575-591,2013 IF 1.8
- 7 M.A.Zakaria, **H.Zamzuri**, S.A.Mazlan, S.M.H.Fahami,' A Path Tracking Algorithm Using Future Prediction Control with Spike Detection for an Autonomous Vehicle Robot', International Journal of Advanced Robotic Systems, Vol. 10, 309:2013 IF 0.8
- 6 Ismail, S.A. Mazlan, **H. Zamzuri** and A.G. Olabi, Fluid-Particle Separation of Magnetorheological Fluid in Squeeze Mode, Japanese Journal of Applied Physics, Vol. 51, 2012, 067301. (IF: 1.018)
- 5 **H. Zamzuri**, A.C. Zolotas, R.M. Goodall and S.A. Mazlan, Advances in Tilt Control Design of High-Speed Railway Vehicles: A Study on Fuzzy Control Methods, International Journal of Innovative Computing, Information and Control, Vol. 8, No. 9, 2012, ISSN: 1349-4198 (IF: 1.664)
- 4 Mohd Rahairi Rani, Hazlina Selamat, **Hairi Zamzuri** and Zuwairie Ibrahim, 'Multi-objective optimization for PID controller Tuning using the global ranking genetic algorithm', Int Journal of Innovative Computing, Information and Control (IJICIC), Vol(8) Number 1(A), 2012, 1349-4198 (IF 2.9)
- 3 A. Mazlan, I. Ismail, **H. Zamzuri** and A.Y.A. Fatah, Compressive and Tensile Stresses of Magnetorheological Fluids in Squeeze Mode, International Journal of Applied Electromagnetics and Mechanics, Vol. 36, 2011, 327-337. (IF: 0.336)
- 2 **Zamzuri, H**, Zolotas, A.C and Goodall, R.M, Tilt control design for high speed trains, A study on multi objective approaches, Vehicle system dynamics, 46(S1): 535-547, 2008 (IF 0.8)
- 1 **Zamzuri, H**, Zolotas, A.C and Goodall, R.M, Intelligent control approaches for tilting railway vehicles, Vehicle system dynamics, 44(S1): 834-843, 2006 (IF 0.8)

5.6 Scopus Journal Publication

- 6 SMH Fahami, H Zamzuri, SA Mazlan, Development of Estimation Force Feedback Control Algorithm for driver steering feel in vehicle steer by wire system: Hardware in the Loop, International Journal of Vehicular Technology, Vol:2015, 1-17, 2015
- 5 Zulkarnain, N., **Zamzuri, H.**, Sam, Y.M., Mazlan, S.A., Zainal, S.M.H.F. Improving vehicle ride and

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- handling using LQG CNF fusion control strategy for an active antiroll bar system, Abstract and Applied Analysis, 2014, 698195
- 4 N. Zulkarnain, **H. Zamzuri**, Y.M. Sam, S.A. Mazlan and S.M.H.F. Zainal, Ride and Handling Analysis for an Active Anti-Roll Bar: Case Study on Composite Nonlinear Control Strategy, International Journal of Automotive and Mechanical Engineering, Vol. 10, 2014, 2122-2143
 - 3 F. Ahmad, S.A. Mazlan, **H. Zamzuri**, H. Jamaluddin, K. Hudha and M. Short, Modelling and Validation of the Vehicle Longitudinal Model, International Journal of Automotive and Mechanical Engineering, Vol. 10, 2014, 2042-2056.
 - 2 S.M.H.Fahami, **H Zamzuri**, S.A.Mazlan and N.Zulkarnain, 'The Design of Vehicle Active Front Steering based on Steer by Wire System, Advance Science Letters 19(1) 2013
 - 1 **Zamzuri, H**, Zolotas, A.C and Goodall, R.M, ' Integral Sliding Mode Control Design for High Speed Train' International Journal of Simulation, System, Science and Technology (IJSSST), Vol 11(4), Sept 2010

5.7 Index Publications

- 30 M.H.M. Ariff, **H. Zamzuri**, N.R.N. Idris, S.A. Mazlan, M.A.M. Nordin, Direct Yaw Moment Control of Independent-Wheel-Drive Electric Vehicle (IWD-EV) via Composite Nonlinear Controller, Proceedings of the First International Conference on Systems Informatics, Modelling and Simulation (SIMS '14), 2014, 112-117.
- 29 D. Derawi, N.D. Salim, **H. Zamzuri**, H. Liu, M.A.A. Rahman and S.A. Mazlan, Robust Attitude Controller for Uncertain Hexarotor Micro Aerial Vehicles (MAVs), 2014 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2014), 2014, 4776-4781.
- 28 M.H.M. Ariff, **H. Zamzuri**, N.R.N. Idris, S.A. Mazlan and M.A.M. Nordin, Independent Torque Control of an Independent-Wheel-Drive Electric Vehicle, Applied Mechanics and Materials, Vol. 663, 2014, 493-497.
- 27 M.A. Zakaria, **H Zamzuri**, R. Mamat and S.A. Mazlan, Optimized Potential Radius Reference Generator Algorithm for Autonomous Vehicle Controller Development, Applied Mechanics and Materials, Vol. 663, 2014, 198-202.
- 26 M. Zeinali, S.A. Mazlan, A.Y.A. Fatah and **H. Zamzuri**, A GA-Weighted Adaptive Neuro-Fuzzy Model to Predict the Behaviour of Magnetorheological Damper, Applied Mechanics and Materials, Vol. 663, 2014, 203-207
- 25 N. Zulkarnain, **H. Zamzuri**, and S.A. Mazlan, LQG Control Design for Vehicle Active Anti-Roll Bar System, Applied Mechanics and Materials, Vol. 663, 2014, 146-151.
- 24 C.Y. Low, H. Zamzuri and **S.A. Mazlan**, Simple Robust Road Lane Detection Algorithm, ICIAS – 5th International Conference on Intelligent and Advanced Systems: Technologies Convergence for Sustainable Future, 2014, 6869550.
- 23 D. Derawi, N.D. Salim, M.A.A. Rahman, **S.A. Mazlan** and H. Zamzuri, Modeling, Attitude Estimation, and Control of Hexarotor Micro Aerial Vehicle (MAV), Proceedings of the IEEE International Conference on Industrial Technology (ICIT), 2014, 55-60.
- 22 N.D. Salim, D. Derawi, S.S. Abdullah, **S.A. Mazlan** and H. Zamzuri, PID plus LQR Attitude Control for Hexarotor MAV in Indoor Environments, Proceedings of the IEEE International Conference on Industrial Technology (ICIT), 2014, 85-90.
- 21 M. Yazid, S.A. Mazlan, T. Kikuchi and **H. Zamzuri**, Experimental Investigation of Multiple Coils Magnetorheological Damper under Dynamic Loadings, Applied Mechanics and Materials, Vol. 660, 2014, 863-867.
- 20 F. Imaduddin, S.A. Mazlan, **H. Zamzuri** and M.A.A. Rahman, Bypass Rotary Magnetorheological Damper for Automotive Applications, Applied Mechanics and Materials, Vol. 663, 2014, 685-689
- 19 L.H. Hamdan, S.A. Mazlan, S. Sarip, **H. Zamzuri** and M.A.A. Rahman, Selection of Materials in Designing Magnetorheological Brake, Applied Mechanics and Materials, Vol. 663, 2014, 700-704.
- 18 N.A.M. Azubir, M.K. Hassan, **H. Zamzuri** and S.A. Mazlan, Development of PROTON Electric Vehicle Control Unit (eVCU) using Machine Deterministic Rule-Based Approach, Applied Mechanics and Materials, Vol. 663, 2014, 532-538.
- 17 A Fauzi, SA Mazlan, **H Zamzuri**, 'Modeling and Validation of Quarter Vehicle Traction Model, Applied Mechanics and Materials 554, 489-493
- 16 N Zulkarnain, **H Zamzuri**, SA Mazlan, 'Combined CNF with LQR in Improving Ride and Handling for Ground Vehicle', Applied Mechanics and Materials 575, 749-752, 2014
- 15 SA Saruchi, **H Zamzuri**, SA Mazlan, SMH Fahami, N Zulkarnain, 'Wheel Synchronization Control in Steer-by-Wire Using Composite Nonlinear Feedback', Applied Mechanics and Materials 575, 762-765: 2014
- 14 Ubaidillah, S.A. Mazlan, J. Sutrisno and **H. Zamzuri**, Potential Applications of Magnetorheological Elastomers, Applied Mechanics and Materials, Vol. 663, 2014, 695-699.
- 13 SMH Fahami, **H Zamzuri**, SA Mazlan, SA Saruchi, 'The Variable Steering Ratio for Vehicle Steer by Wire System Using Hyperbolic Tangent Method', Applied Mechanics and Materials 575, 781-784, 2014
- 12 IM Yazid, S.A. Mazlan, **H Zamzuri**, MJ Mughni, S Chuprat, 'Parameters consideration in designing a magnetorheological damper', Key Engineering Material, Vol 543, 487-490, 2013

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- 11 M.A.Zakaria, **H.Zamzuri**, S.A.Mazlan, S.M.H.Fahami, Vehicle Path Tracking Using Future Prediction Steering Control, *Procedia Engineering*, 2012
 - M.H.M Ariff, **H Zamzuri**, N.R.N Idris, S.A.Mazlan, 'Antilock Braking System Slip Control Modelling Revisited, *Applied Mechanics and Materials*, 393, 637-643, 2013
 - 9 M. Yazid, S.A. Mazlan, **H. Zamzuri**, M.J. Mughni and S. Chuprat, Parameters Consideration in Designing a Magnetorheological Damper, *Key Engineering Materials*
 - 7 I. Ismail, S.A. Mazlan, **H. Zamzuri** and A.G. Olabi, Investigation of Mechanical Performance of Squeezed Magnetorheological Fluid Using Response Surface Method, *Advanced Materials Research*, Vol. 445, 2012, 577-582.
 - 10 S.M.H.Fahami, **H Zamzuri**, S.A.Mazlan and N.Zulkarnain, The Design of Vehicle Active Front Steering based on Steer by Wire System, *International Conference on Advanced in Materials Science and Engineering (AMSE 2012)*, 27-28 Sept 2012
 - 9 S.M.H.Fahami, **H.Zamzuri**, S.A.Amri, M.A.Zakaria, Modeling and Simulation of Vehicle Steer-by-wire System, *IEEE Symposium on Humanities, Science and Engineering Research (SHUSER 2012)*, Kuala Lumpur, Malaysia, 24-27 June 2012
 - 8 M. Yazid, S.A. Mazlan, **H. Zamzuri**, M.J. Mughni and S. Chuprat, Electromagnetic Circuit Simulation and Design Parameters of Magnetorheological Damper, *The International Conference in Materials and Applications for Sensors and Transducers (IC-MAST 2012)*, Budapest, Hungary, May 24-28, 2012.
 - 7 **H. Zamzuri**, S.A. Mazlan, A.C. Zolotas and R.M. Goodall, Verification of Intelligent Fuzzy Mechanism Control Strategy Using 9 DOF Complex Tilting Railway Vehicle Model, *IEEE International Conference on Intelligent Computing and Intelligent Systems (ICIS 2011)*, Guangzhou, China, November 18-20, 2011
 - 6 I. Ismail, S.A. Mazlan, **H. Zamzuri** and A.G. Olabi, Investigation of Mechanical Performance of Squeezed Magnetorheological Fluid Using Response Surface Method, presented at the 14th International Conference on Advances in Materials and Processing Technologies (AMPT 2011), Istanbul, Turkey, July 13-16, 2011
 - 5 Rani, M.R; Selamat H; **Zamzuri, H**, Ahmad, F; PID Controller optimization for a rotational inverts pendulum using genetic algorithm, 4th the International Conference on Modeling, Simulation and Applied Optimization (ICMSAO11), 19-21 April 2011, Kuala Lumpur
 - 4 Ramli, M.S.A. , **Zamzuri, H** , Abidin, M.S.Z., Tracking human movement in office environment using video processing, 4th the International Conference on Modeling, Simulation and Applied Optimization (ICMSAO11), 19-21 April 2011, Kuala Lumpur
 - 3 **Zamzuri, H**, Zolotas, A.C and Goodall, R.M, ' Integral Sliding Mode Control Design for High Speed Train', presented at Conference on Instrumentation and Mechatronic (CIM 09), Jun 2009, Melacca, Malaysia
 - 2 **Zamzuri, H**, Zolotas, A.C and Goodall, R.M, LQG with fuzzy correction mechanism in titling railway vehicle control design, presented at 3rd IFAC workshop on Advanced Fuzzy and Neural Network (AFNC07), Valenciennes, France, 29-30 Oct., 2007
 - 1 **Zamzuri, H**, Zolotas, A.C and Goodall, R.M, Optimized Intelligent tilt controller scheme using genetic algorithm, UKACC International Control Conference 2006, Glasgow, 30th August-1st September 2006, pp 1-6, ISSN 0 947649549

5.5 Article evaluation in Journal

- | | |
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| 2015 | International Journal of Advanced Robotic Systems, 'Time Variant Predictive Control of Autonomous vehicles' |
| 2013 | International Journal of Simulation: System, Science and Technology (IJSST) Title : <i>Road vehicle following system with adaptive controller gain using Model Reference Adaptive Control Method</i> |
| 2014 | America Journal of Intelligent System, 'NNARMAX Model Identification of an AS-WWTP using ARLS: Part 2 Dynamic Modeling of the secondary Settler and Clarifier' |
| 2014 | Review for Jurnal Teknologi Special Edition (Artificial Intelligence and Robotic) |
| 2014 | Transactions on Intelligent Transportation System, <i>H-Infinity control approach to the problem of autonomous vehicle navigation</i> |

PART 6 : CONTRIBUTION TO THE UNIVERSITY'S ADMINISTRATION OR COMMUNITY

6.1 Additional responsibilities

(Vice-Chancellor, Dean, Director, Deputy Dean, Assistant Director, Principal of Students' Dormitories, Department Head, Course Head/ Panel, Laboratory Head, Coordinator or Students' Dormitories Administrator)

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- 2015-2016 : MJIT Technical Tender Committee
 - 2011 -2013 : Head of Research Lab: UTM-Proton Active safety
 - 2010 -2011 : UTM Razak School Academic Faculty Committee member
 - 2008-2010 : Head of Microprocessor Lab
 - 2009-2010 : Head of System Lab
 - 2002-2004 : SPACE coordinator for KST

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2002-2004 : Head of Control Lab for KST  
2001-2004 : Subject coordinator for DDB subjects

#### **6.2 Administrative Duties**

2011-2012 : Head of Research Lab: UTM-Proton Active Safety.  
2010-2010 : IT Manager for UTM Razak School of Engineering and Advanced Tech.

#### **6.3 Member of other associations in UTM**

2012 : Working committee for Festival Inovasi and Kreativiti (UTMOST IMAGINATION)  
2010-2011 : Member for UTM Razak School Technical Evaluation Quotation Binding  
2008-2010 : Member of E-learning at College of Science and Technology  
2008-2009 : Member of coordinator for Penerbitan Ilmiah KST  
2008-2009 : Member for Journal Teknologi (JUTEK) for KST  
2008-2009 : Panel for Post Graduate Committee for KST  
2003-2005 : Committee member for Diploma Technology Curriculum and Syllabus development

#### **6.4 Activities**

Panel Evaluation for UTM Razak School Postgraduate Research Colloquium  
Assessor for 19th Colloquium For Engineering Doctorate Students 2011/2012

#### **6.5 Editorial board membership**

- 1 Appointed as an editorial board member of the American Journal of Intelligent Systems under Scientific & Academic Publishing for 10/12/2011 - 09/12/2012.
- 2 Appointed as Editorial Panel for Engineering Division Internal Journal , Proton Sdn Bhd from August 2011 - 2012, 2014-2015
- 3 Appointed as Editorial member for Central European Journal of Engineering (ISBN: 1896-1541)

#### **6.6 Other activities outside the university (talk/ academic forum/ non-academic)**

- 1 Lecture at the Research Methodology Programme for Kolej Komuniti. 10-12/09/2011.
- 2 Moderator at Automotive Safety Week 2012, Melaka organised by Malaysia Institute of Road Safety Research (MIROS), 22-25th May 2012
- 3 Chairperson for Technical Session at 4th International Conference on Modeling, Simulation and Applied Optimization 2011 (ICMSAO'11) 19-21st April 2011

#### **6.7 Sports**

1. 2009. Represented College of Science and Technology, UTM, KL in Bowling and Chess tournaments (Inter Department)

### **PART 7 : AWARDS/ PRIZES**

*(Please state awards/ prizes won for any distinguished academic achievement and contribution to the community)*

- 3 Silver Model, Malaysia Technology Expo, Intelligent Urban Shuttle, 2012
- 2 Silver Award, Malaysia Technology Expo, Hybrid Magnetorheological Damper
- 1 Gold Award, Malaysia Technology Expo, Compact Magnetorheological Valve, 2012

### **PART 8 : PROFESSIONAL MEMBERSHIPS**

Member of Board of Engineering Malaysia  
Member of Institute of Electrical Electronic Engineering (IEEE)  
Member of IEEE Robotic and Automation Society (RAS)